

SMART SIGN LANGUAGE GLOVES



Startup Name: INTELLIX TITANS

We provide an offline, app-free wearable that translates sign language into spoken audio in real-time.

**Giving a voice
to the
voiceless,
instantly.**

Problem & Market Size

EUREKA 2026

The Problem



The new born deaf babies and general public doesn't know sign language, causing daily communication friction. Current assistive gloves use fragile flex sensors and require bulky mobile apps.

Speech & hearing-impaired individuals, specialized schools, and medical facilities.



Target Customers

TAM (Total Addressable Market)

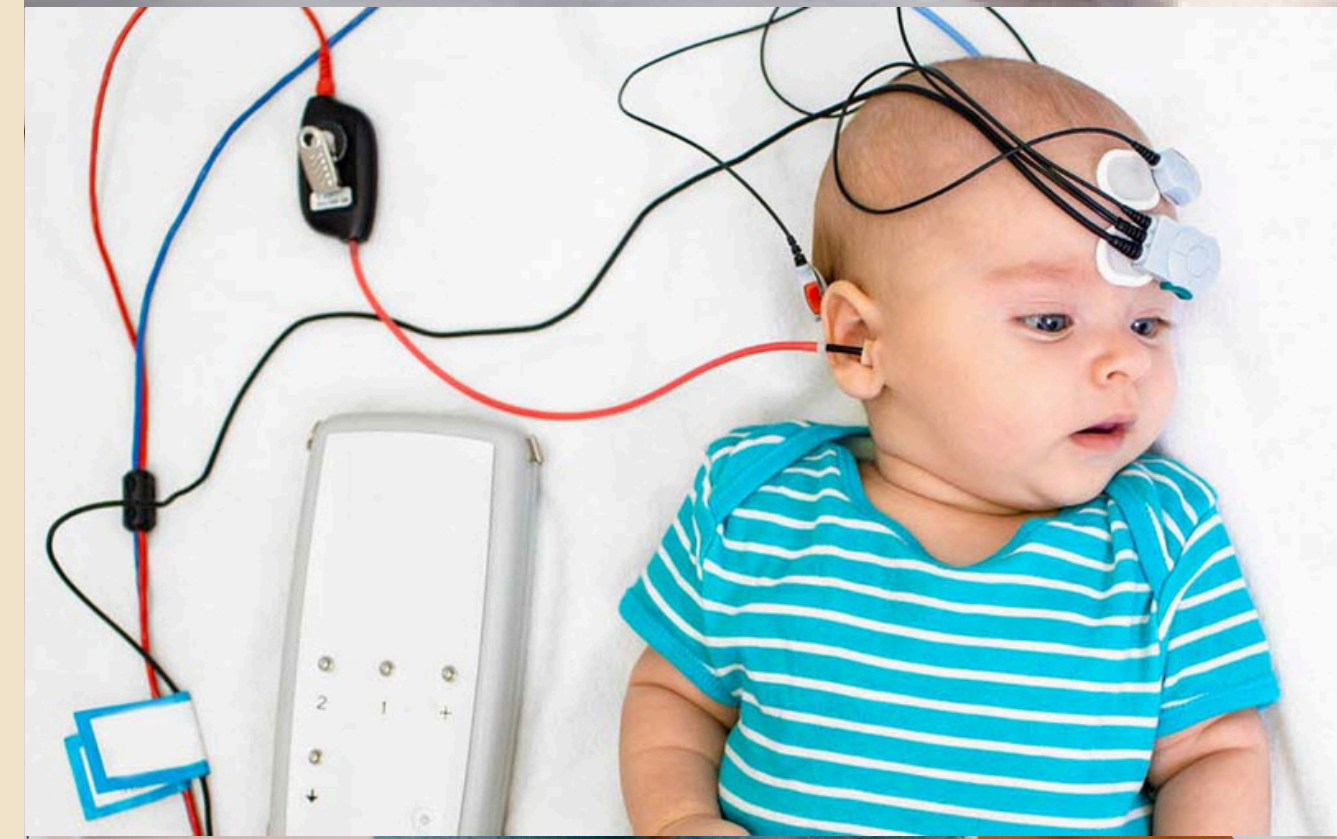


430 Million people globally with disabling hearing/speech loss (WHO stat).

~5 Million individuals in India relying on sign language daily.



SAM (Serviceable Available Market)



Solution & Product Demo

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Proposed Solution

A low-cost wearable utilizing a 5-bit binary touch system and edge-computing.

A edge-computing wearable that acts as its own offline server to translate sign language seamlessly.



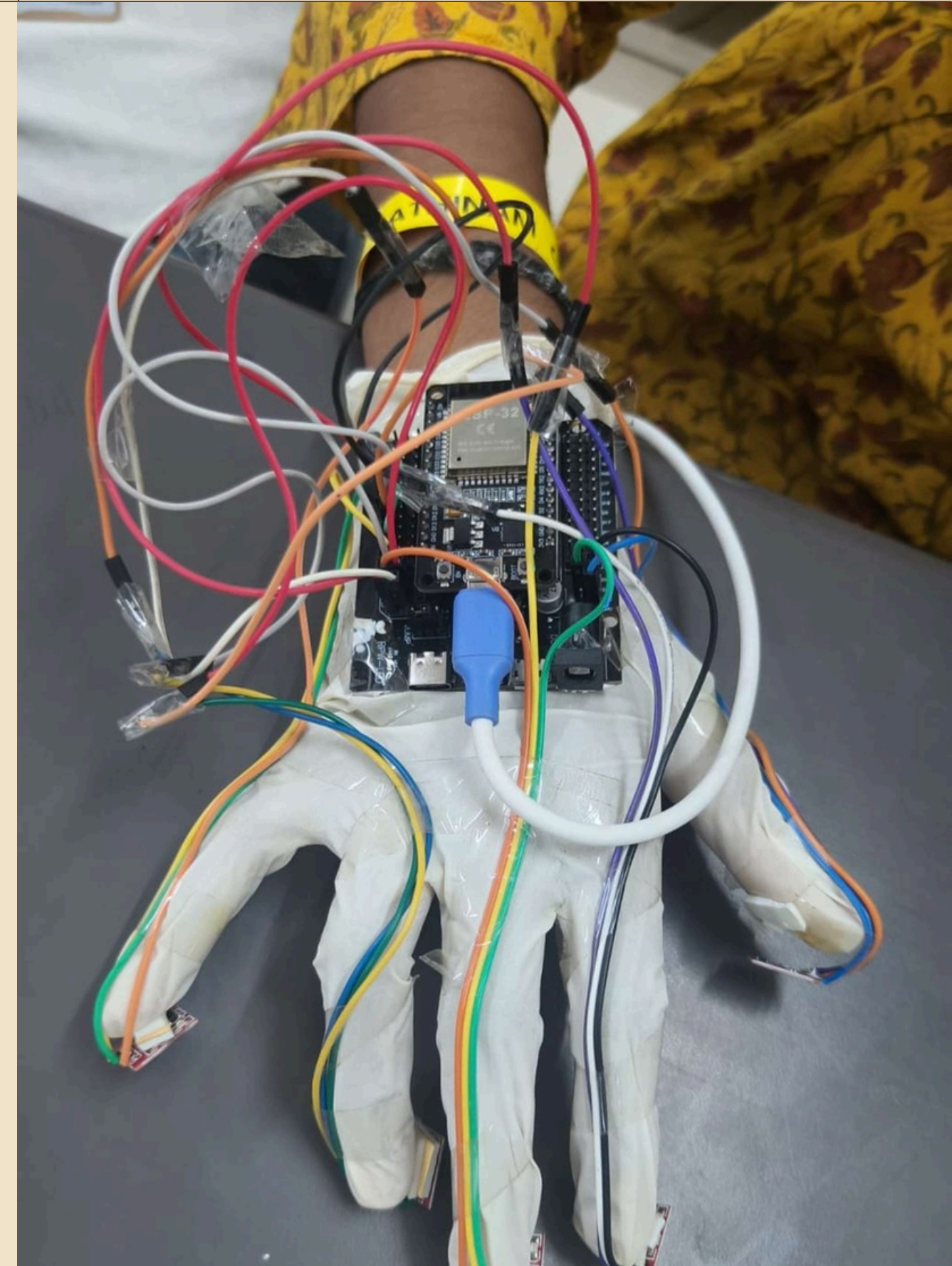
Zero-Install: Hosts its own local web server; no internet or app downloads needed.

Key Features:

Durable & Cheap: Replaces expensive flex sensors with capacitive touch pads.

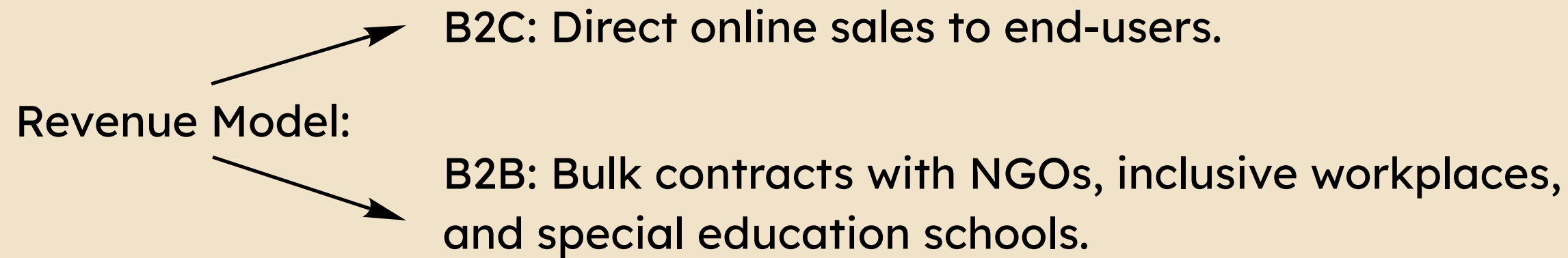
Anti-Misfire: MPU6050 motion tracking verifies intentional hand swipes.

Prototype



Business Model & Traction

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Pricing Strategy: Build cost is under ₹750. Target retail price: ₹2,500 (Highly accessible).

MVP & Traction: We have a fully functional Minimum Viable Product (MVP) operating smoothly offline, ready for real-world user beta testing.

We have successfully built a fully functional MVP that validates our 95% cost-reduction hypothesis.

- App-free concept proven flawlessly
- Eliminating friction from expensive competitor gloves



OVERVIEW AND FLOWGRAPH

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EDGE COMPUTING & LOGIC

- 5-Bit Binary Matrix
- Sensor Fusion (Anti-Misfire)
- Offline Local Web Server

3D VISUALIZATION DASHBOARD

- Three.js Rendering
- Live Telemetry
- Dynamic Lip-Sync

APP-FREE COMMUNICATION

- No App Downloads
- Async WebSockets
- Native TTS Audio



MULTI-SENSOR WEARABLE SUITE

- Capacitive Touch (TTP223)
- Motion Tracking (MPU6050)
- ESP32-C3 Microcontroller
- Power Management

OPERATION WORKFLOW

1. Power On:Glove creates WiFi
2. Connect:Open browser on phone
3. Form Sign:Touch sensors activate
4. Swipe:Motion triggers WebSocket
5. Output:Avatar Speaks & Animates



TEAM

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John Stewartsson J R - Hardware/Firmware Development

Joshua Miracle J - Sensor Integration and Testing

Mahalakshmi P - HTML,CSS,JAVA Coder

Narmathakavi C S - 3D Integration and Hardware Design

Funding Required: Seeking ₹50,000 seed grant.

Use of Funds

- PCB miniaturization
- Ergonomic glove materials
- Funding a 20-user pilot program

Next Milestone

Finalizing the commercial-ready prototype for mass manufacturing.



Thank you for listening!

By: IntelliX Titan's Team

